

# 230941\_\_RAJ\_\_SAURAV

## Assignment 1

2025-09-10

### Question 2.1

#### Ques.2.1(a)

```
library(astsa)
```

```
## Warning: package 'astsa' was built under R version 4.4.3
```

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.4.3
```

```
trend <- time(jj) - 1970
xt <- log(jj)
Q <- factor(cycle(jj))
reg <- lm(xt ~ 0 + trend + Q)
model.matrix(reg)
```

```
##      trend Q1 Q2 Q3 Q4
## 1 -10.00  1  0  0  0
## 2  -9.75  0  1  0  0
## 3  -9.50  0  0  1  0
## 4  -9.25  0  0  0  1
## 5  -9.00  1  0  0  0
## 6  -8.75  0  1  0  0
## 7  -8.50  0  0  1  0
## 8  -8.25  0  0  0  1
## 9  -8.00  1  0  0  0
## 10 -7.75  0  1  0  0
## 11 -7.50  0  0  1  0
## 12 -7.25  0  0  0  1
## 13 -7.00  1  0  0  0
## 14 -6.75  0  1  0  0
## 15 -6.50  0  0  1  0
## 16 -6.25  0  0  0  1
## 17 -6.00  1  0  0  0
## 18 -5.75  0  1  0  0
## 19 -5.50  0  0  1  0
```

```

## 20 -5.25 0 0 0 1
## 21 -5.00 1 0 0 0
## 22 -4.75 0 1 0 0
## 23 -4.50 0 0 1 0
## 24 -4.25 0 0 0 1
## 25 -4.00 1 0 0 0
## 26 -3.75 0 1 0 0
## 27 -3.50 0 0 1 0
## 28 -3.25 0 0 0 1
## 29 -3.00 1 0 0 0
## 30 -2.75 0 1 0 0
## 31 -2.50 0 0 1 0
## 32 -2.25 0 0 0 1
## 33 -2.00 1 0 0 0
## 34 -1.75 0 1 0 0
## 35 -1.50 0 0 1 0
## 36 -1.25 0 0 0 1
## 37 -1.00 1 0 0 0
## 38 -0.75 0 1 0 0
## 39 -0.50 0 0 1 0
## 40 -0.25 0 0 0 1
## 41 0.00 1 0 0 0
## 42 0.25 0 1 0 0
## 43 0.50 0 0 1 0
## 44 0.75 0 0 0 1
## 45 1.00 1 0 0 0
## 46 1.25 0 1 0 0
## 47 1.50 0 0 1 0
## 48 1.75 0 0 0 1
## 49 2.00 1 0 0 0
## 50 2.25 0 1 0 0
## 51 2.50 0 0 1 0
## 52 2.75 0 0 0 1
## 53 3.00 1 0 0 0
## 54 3.25 0 1 0 0
## 55 3.50 0 0 1 0
## 56 3.75 0 0 0 1
## 57 4.00 1 0 0 0
## 58 4.25 0 1 0 0
## 59 4.50 0 0 1 0
## 60 4.75 0 0 0 1
## 61 5.00 1 0 0 0
## 62 5.25 0 1 0 0
## 63 5.50 0 0 1 0
## 64 5.75 0 0 0 1
## 65 6.00 1 0 0 0
## 66 6.25 0 1 0 0
## 67 6.50 0 0 1 0
## 68 6.75 0 0 0 1
## 69 7.00 1 0 0 0
## 70 7.25 0 1 0 0
## 71 7.50 0 0 1 0
## 72 7.75 0 0 0 1
## 73 8.00 1 0 0 0

```

```
## 74 8.25 0 1 0 0
## 75 8.50 0 0 1 0
## 76 8.75 0 0 0 1
## 77 9.00 1 0 0 0
## 78 9.25 0 1 0 0
## 79 9.50 0 0 1 0
## 80 9.75 0 0 0 1
## 81 10.00 1 0 0 0
## 82 10.25 0 1 0 0
## 83 10.50 0 0 1 0
## 84 10.75 0 0 0 1
## attr("assign")
## [1] 1 2 2 2 2
## attr("contrasts")
## attr("contrasts")$Q
## [1] "contr.treatment"
```

```
summary(reg)
```

```
##
## Call:
## lm(formula = xt ~ 0 + trend + Q)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.29318 -0.09062 -0.01180  0.08460  0.27644
##
## Coefficients:
##      Estimate Std. Error t value Pr(>|t|)
## trend 0.167172   0.002259   74.00  <2e-16 ***
## Q1     1.052793   0.027359   38.48  <2e-16 ***
## Q2     1.080916   0.027365   39.50  <2e-16 ***
## Q3     1.151024   0.027383   42.03  <2e-16 ***
## Q4     0.882266   0.027412   32.19  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1254 on 79 degrees of freedom
## Multiple R-squared:  0.9935, Adjusted R-squared:  0.9931
## F-statistic: 2407 on 5 and 79 DF, p-value: < 2.2e-16
```

### Ques.2.1(b)

Estimated average annual Increase in log(EPS) is: 0.167

```
avg_annual_increase <- coef(reg)["trend"]
cat("Average annual Increase in log(EPS) is: ", avg_annual_increase, "\n")
```

```
## Average annual Increase in log(EPS) is: 0.1671722
```

### Ques.2.1(c)

The average logged earnings rate will decrease by 23.57% as we go from the third quarter to the fourth quarter

```
log_eps_Q3 <- coef(reg)["Q3"]
log_eps_Q4 <- coef(reg)["Q4"]
change <- (log_eps_Q4 - log_eps_Q3)
percent_dec <- (exp(change) - 1) * 100

cat("Average logged earning rate decreases from the Q3 to Q4 by:", abs(percent_dec), "%")
```

```
## Average logged earning rate decreases from the Q3 to Q4 by: 23.56716 %
```

### Ques.2.1(d)

If we include the intercept then the model automatically removed average logged ERP of Q1 and taken Q2, Q3, Q4 as difference from  $\beta_0$  or Q1. trend coefficient still remains the same.

```
# add intercept in the model
reg2 <- lm(log(jj) ~ trend + Q)
summary(reg2)
```

```
##
## Call:
## lm(formula = log(jj) ~ trend + Q)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.29318 -0.09062 -0.01180  0.08460  0.27644
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.052793   0.027359  38.480 < 2e-16 ***
## trend        0.167172   0.002259  73.999 < 2e-16 ***
## Q2           0.028123   0.038696   0.727  0.4695
## Q3           0.098231   0.038708   2.538  0.0131 *
## Q4          -0.170527   0.038729  -4.403 3.31e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1254 on 79 degrees of freedom
## Multiple R-squared:  0.9859, Adjusted R-squared:  0.9852
## F-statistic: 1379 on 4 and 79 DF, p-value: < 2.2e-16
```

We can not include all four quarterly dummies Q1, Q2, Q3 and Q4 along with an intercept because of perfect multicollinearity:

$$Q1 + Q2 + Q3 + Q4 = 1$$

so if we include intercept then sum of the dummies equals the intercept that is the intercept is exactly a linear combination of the dummies, so including all of them would make the design matrix singular.

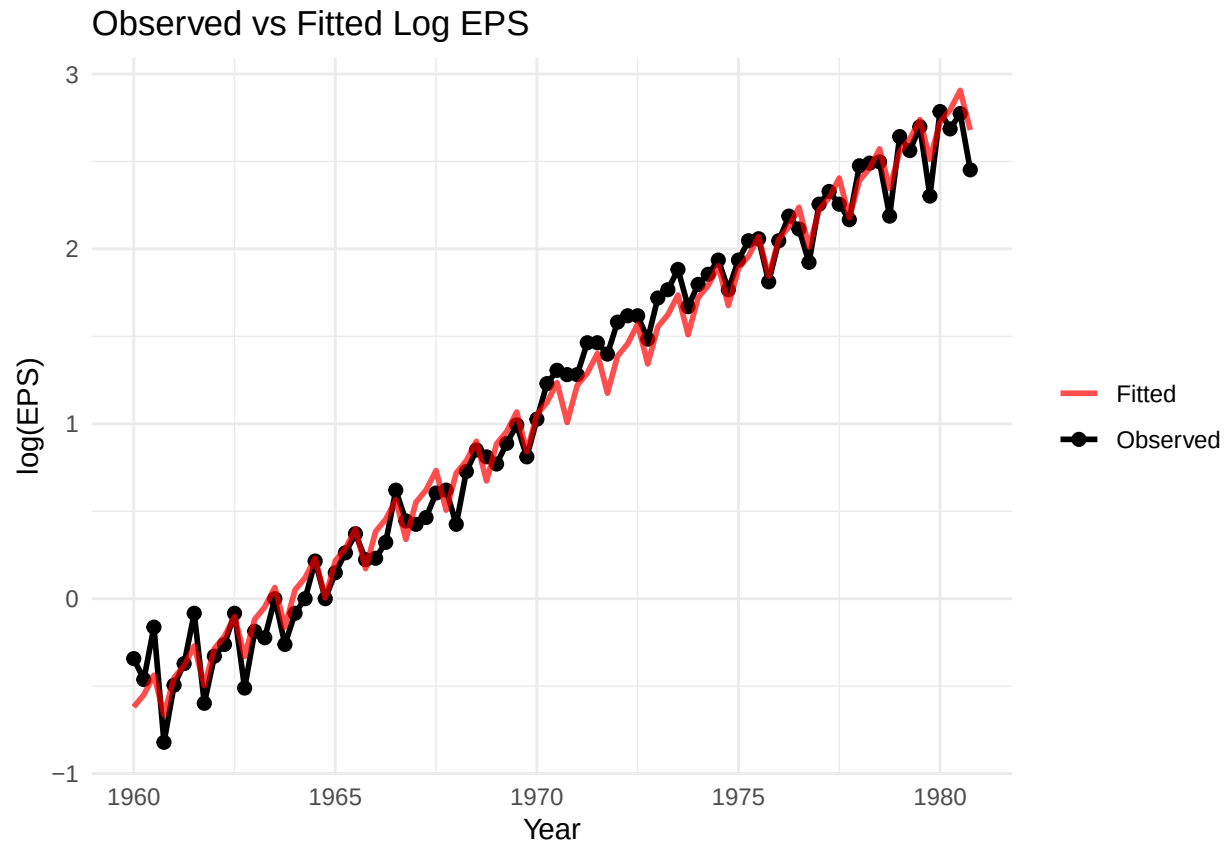
R handled it by automatically dropping one dummy (Q1) which serves as the baseline quarters.

Ques.2.1(e)

```
df <- data.frame(  
  t = time(jj),  
  x_t = xt,  
  x_hat_t = fitted(reg)  
)  
  
ggplot(df, aes(x = t))+  
  geom_line(aes(y = x_t, color = "Observed"), size=1)+  
  geom_point(aes(y = x_t, color="Observed"), size=2) +  
  geom_line(aes(y = x_hat_t, color = "Fitted"), size = 1, alpha = 0.7)+  
  labs(y = "log(EPS)", x = "Year", title = "Observed vs Fitted Log EPS") +  
  scale_color_manual(values=c("Observed"="black", "Fitted"="red")) +  
  theme_minimal() +  
  theme(legend.title = element_blank())
```

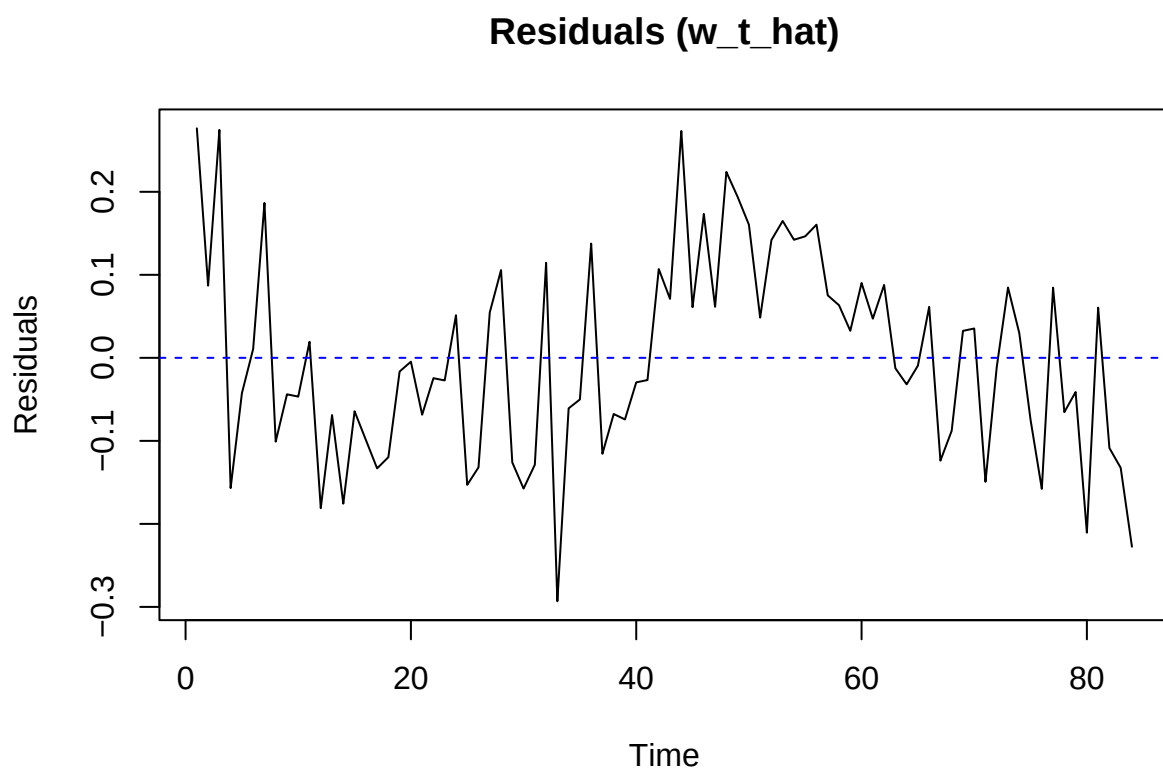
```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.  
## i Please use 'linewidth' instead.  
## This warning is displayed once every 8 hours.  
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was  
## generated.
```

```
## Don't know how to automatically pick scale for object of type <ts>. Defaulting  
## to continuous.  
## Don't know how to automatically pick scale for object of type <ts>. Defaulting  
## to continuous.
```



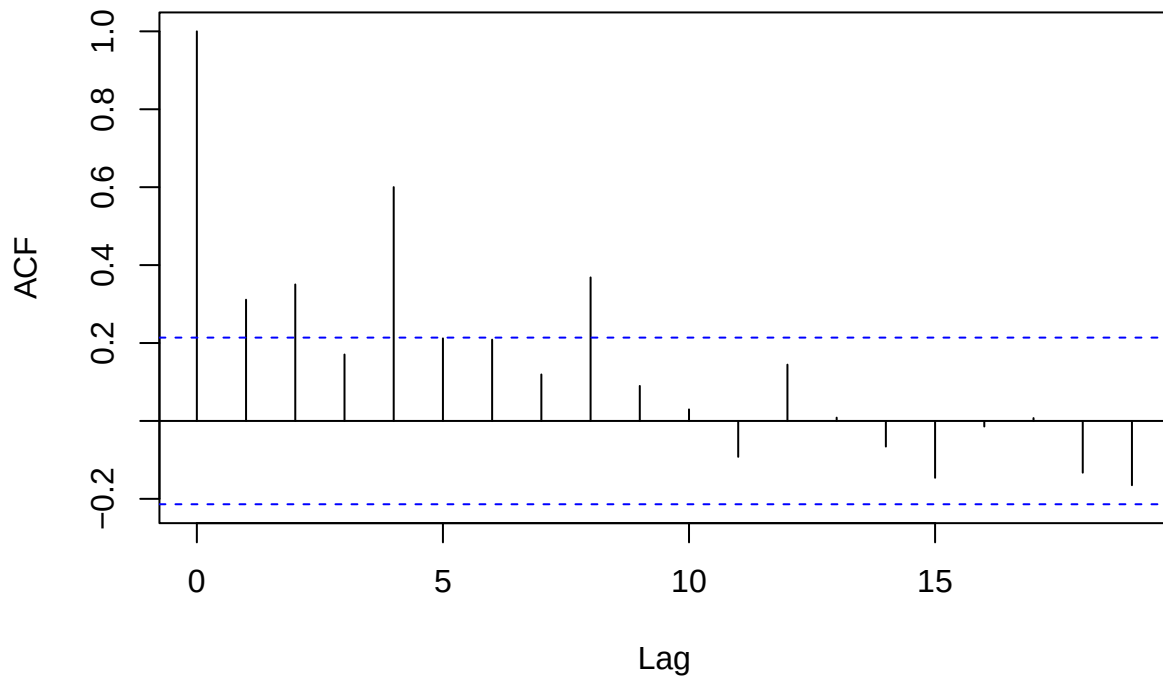
Examine the Residual ( $x_t - \hat{x}_t$ )

```
wt_hat <- resid(reg)
#plot
ts.plot(wt_hat, ylab="Residuals", main="Residuals (w_t_hat)")
abline(h=0, col="blue", lty=2)
```



```
# ACF  
acf(wt_hat, main="ACF of Residuals")
```

## ACF of Residuals



Conclusion: The ACF shows a large positive lag-1 spike (goes outside confidence interval band) and significant peaks at seasonal lags (multiples of 4), with a slow decay. This indicates remaining serial and seasonal dependence.

The model fits the data well in terms of R-Squared and significance, but the residuals are not white since the ACF shows strong autocorrelation.

## Question 2.2

### Ques.2.2(a)

```
# Data Import
ctemp <- tempr - mean(tempr)
temp2 <- ctemp^2
trend <- time(cmort)

part_lag4 <- ts(c(rep(NA, 4), head(as.numeric(part), -4)), start = start(part), frequency = frequency(p

data_lagged_ts <- ts.intersect(
  cmort = cmort,
  trend = trend,
  ctemp = ctemp,
  temp2 = temp2,
  part = part,
  part_lag4 = part_lag4
```

```
)

data_lagged_df <- na.omit(as.data.frame(data_lagged_ts))

fit_old <- lm(cmort ~ trend + ctemp + temp2 + part, data = data_lagged_df)
fit_new <- lm(cmort ~ trend + ctemp + temp2 + part + part_lag4, data = data_lagged_df)

summary(fit_old)
```

```
##
## Call:
## lm(formula = cmort ~ trend + ctemp + temp2 + part, data = data_lagged_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -19.1296  -4.2181  -0.4848   3.7702  29.1977
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.843e+03  2.019e+02  14.082 < 2e-16 ***
## trend        -1.402e+00  1.022e-01 -13.717 < 2e-16 ***
## ctemp        -4.741e-01  3.170e-02 -14.958 < 2e-16 ***
## temp2         2.268e-02  2.834e-03   8.001 8.7e-15 ***
## part          2.564e-01  1.892e-02  13.549 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.389 on 499 degrees of freedom
## Multiple R-squared:  0.5945, Adjusted R-squared:  0.5912
## F-statistic: 182.9 on 4 and 499 DF,  p-value: < 2.2e-16
```

```
summary(fit_new)
```

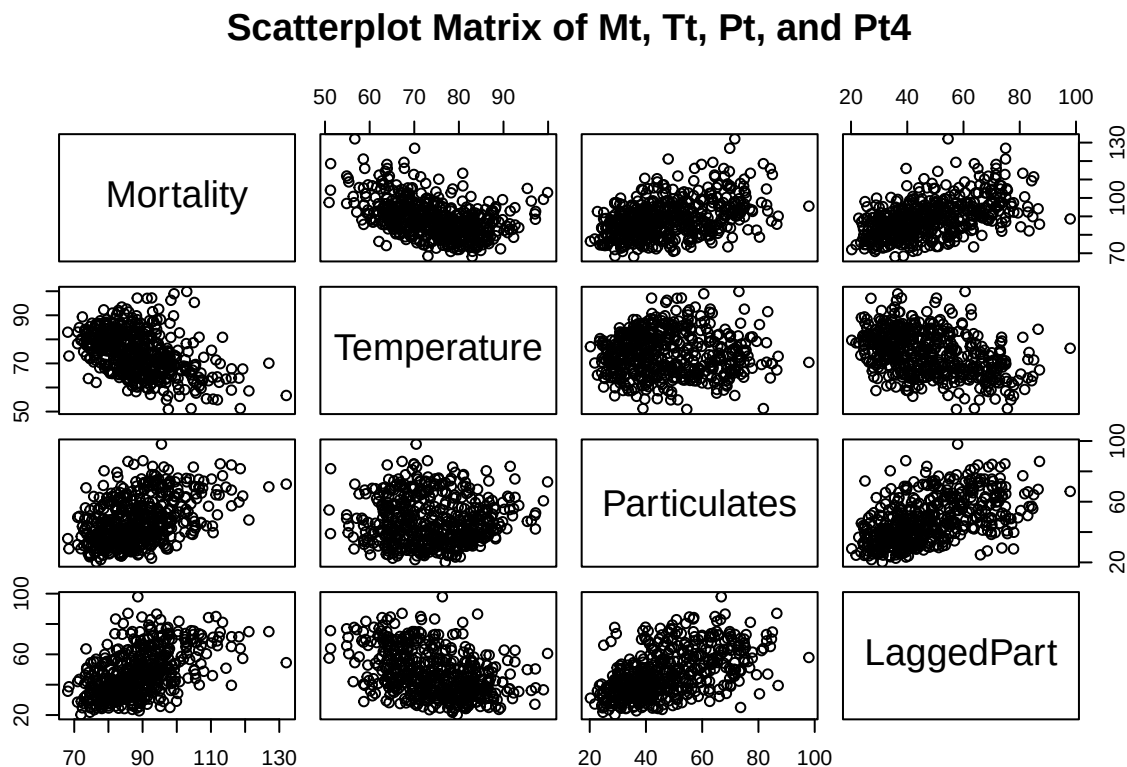
```
##
## Call:
## lm(formula = cmort ~ trend + ctemp + temp2 + part + part_lag4,
##     data = data_lagged_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -18.228  -4.314  -0.614   3.713  27.800
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.808e+03  1.989e+02  14.123 < 2e-16 ***
## trend        -1.385e+00  1.006e-01 -13.765 < 2e-16 ***
## ctemp        -4.058e-01  3.528e-02 -11.503 < 2e-16 ***
## temp2         2.155e-02  2.803e-03   7.688 8.02e-14 ***
## part          2.029e-01  2.266e-02   8.954 < 2e-16 ***
## part_lag4     1.030e-01  2.485e-02   4.147 3.96e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 6.287 on 498 degrees of freedom
## Multiple R-squared:  0.608, Adjusted R-squared:  0.6041
## F-statistic: 154.5 on 5 and 498 DF,  p-value: < 2.2e-16
```

Conclusion: Four-week lag of particulates (part\_lag4) shows a positive effect on cmort with an estimate of 0.103 (SE = 0.0248, t Value= 4.147,  $p < 0.001$ ). R-squared slightly increases from 0.594 to 0.604 and residual error decreases, indicating a modest improvement in model fit. Hence, past pollution contributes measurably to current mortality, highlighting the importance of considering lagged exposures.

### Ques.2.2(b)

```
# scatter plot
pairs(cbind(
  Mortality=cmort,
  Temperature=tempr,
  Particulates=part,
  LaggedPart = part_lag4),
  main = "Scatterplot Matrix of Mt, Tt, Pt, and Pt4")
```



PairWise Correlations:

```
data <- data.frame(
  Mt = data_lagged_df$cmort,
  Tt = data_lagged_df$ctemp + mean(tempr),
  Pt = data_lagged_df$part,
  Pt4 = data_lagged_df$part_lag4
)

cor_matrix <- cor(data)
print(cor_matrix)
```

```
##           Mt           Tt           Pt           Pt4
## Mt    1.0000000 -0.4369648  0.4422896  0.5209993
## Tt   -0.4369648  1.0000000 -0.0148241 -0.3990848
## Pt    0.4422896 -0.0148241  1.0000000  0.5340505
## Pt4   0.5209993 -0.3990848  0.5340505  1.0000000
```

Mt is more strongly correlated with Pt4 than with Pt suggesting a stronger association with pt4. Both correlations are moderate and positive, so a meaningful association exists but much variation remains unexplained.

Pt and Pt4 are themselves correlated ( $r = 0.534$ ), so including both in a regression may cause multicollinearity.

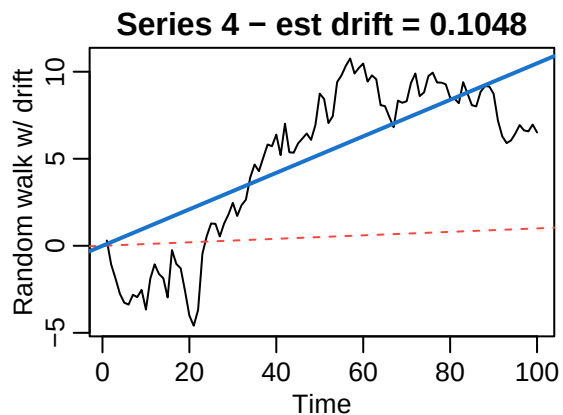
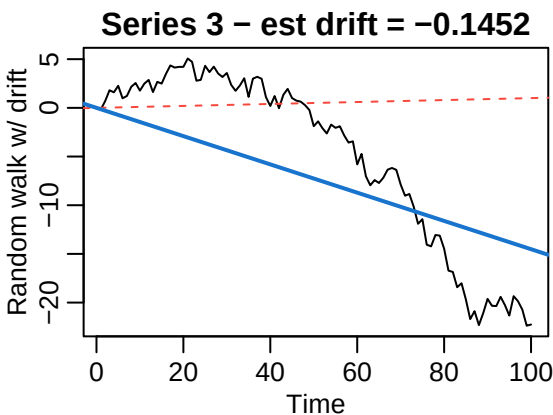
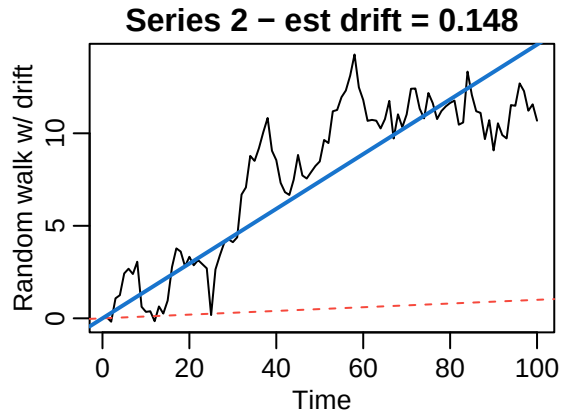
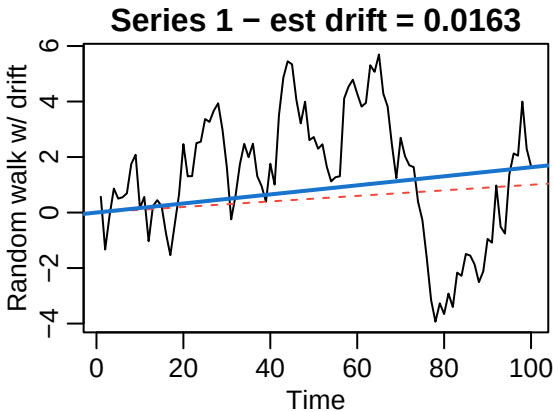
## Question 2.3

```
set.seed(154)
mu <- 0.01
sigma <- 1
n <- 100
time_vec <- 1:n

par(mfrow= c(2, 2), mar = c(2.5, 2.5, 1, 1.5)+ 0.5, mgp = c(1.6, .6, 0))

estimates <- numeric(4)
for(i in 1:4) {
  x <- ts(cumsum(rnorm(n, mean = mu, sd = sigma)))
  regx <- lm(x ~ 0 + time(x))
  estimates[i] <- coef(regx)[1]

  plot(x, ylab = "Random walk w/ drift", xlab = "Time")
  abline(a = 0, b = mu, col = 2, lty = 2)
  abline(regx, col = 4, lwd = 2)
  title(main = paste("Series", i, "- est drift =", round(estimates[i], 4)))
}
```



```
print(estimated)
```

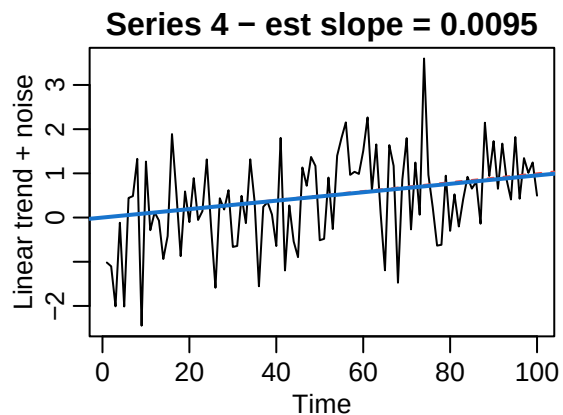
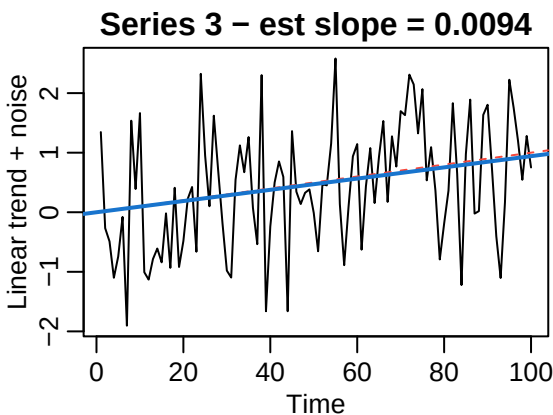
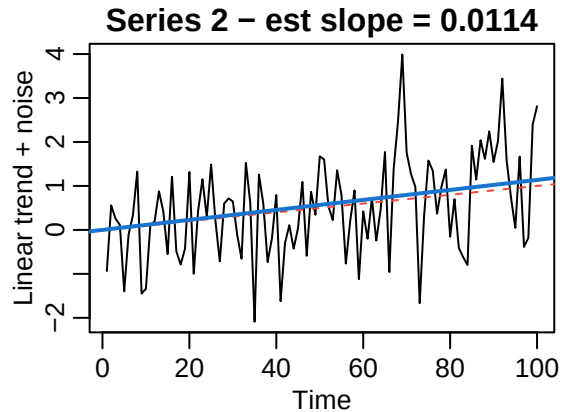
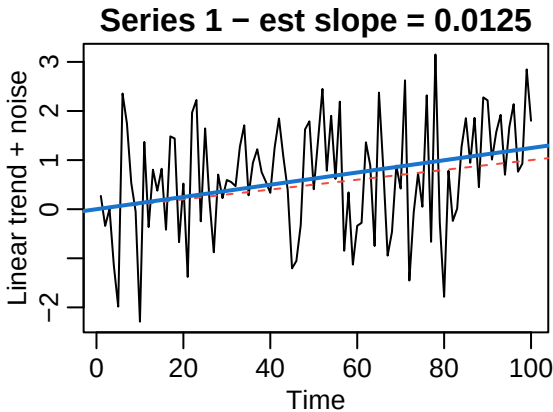
```
## [1] 0.01633042 0.14796015 -0.14517664 0.10483335
```

**Ques.2.3(b)**

```
par(mfrow = c(2, 2), mar = c(2.5, 2.5, 1, 1.5)+0.5, mgp = c(1.6, 0.6, 0))

estimated_b <- numeric(4)

for(i in 1:4){
  w_b <- rnorm(n, mean = 0, sd = sigma)
  y <- mu * time_vec + w_b
  regy <- lm(y ~ 0 + time_vec)
  estimated_b[i] <- coef(regy)[1]
  plot(time_vec, y, type = "l", ylab = "Linear trend + noise", xlab = "Time")
  abline(a = 0, b = mu, col = 2, lty = 2)
  abline(regy, col = 4, lwd = 2)
  title(main = paste("Series", i, "- est slope =", round(estimated_b[i],4)))
}
```



```
print(estimated_b)
```

```
## [1] 0.012462472 0.011371308 0.009414977 0.009536643
```

### Ques.2.3(c)

From this assignment, I learned how to estimate trends and seasonal effects using regression with and without intercepts, and how R handles multicollinearity by dropping one dummy variable. I observed that mortality and EPS data can show meaningful patterns over time, but residual analysis is essential to check model adequacy. The random walk and linear trend simulations helped me understand drift, fitted lines, and the effect of noise on time series. Overall, I gained confidence in interpreting model coefficients, plotting fitted versus observed values, and assessing residual behavior.